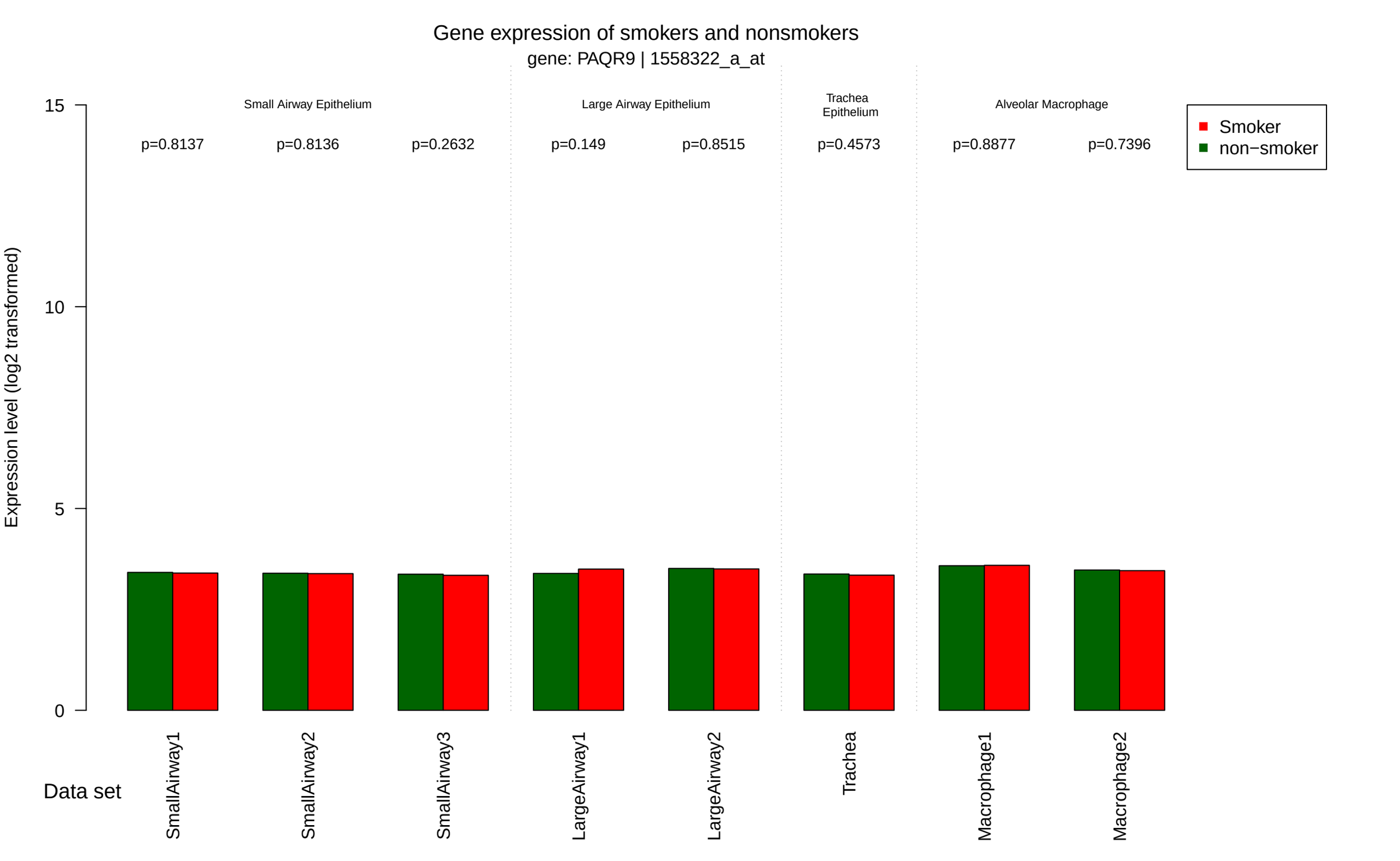
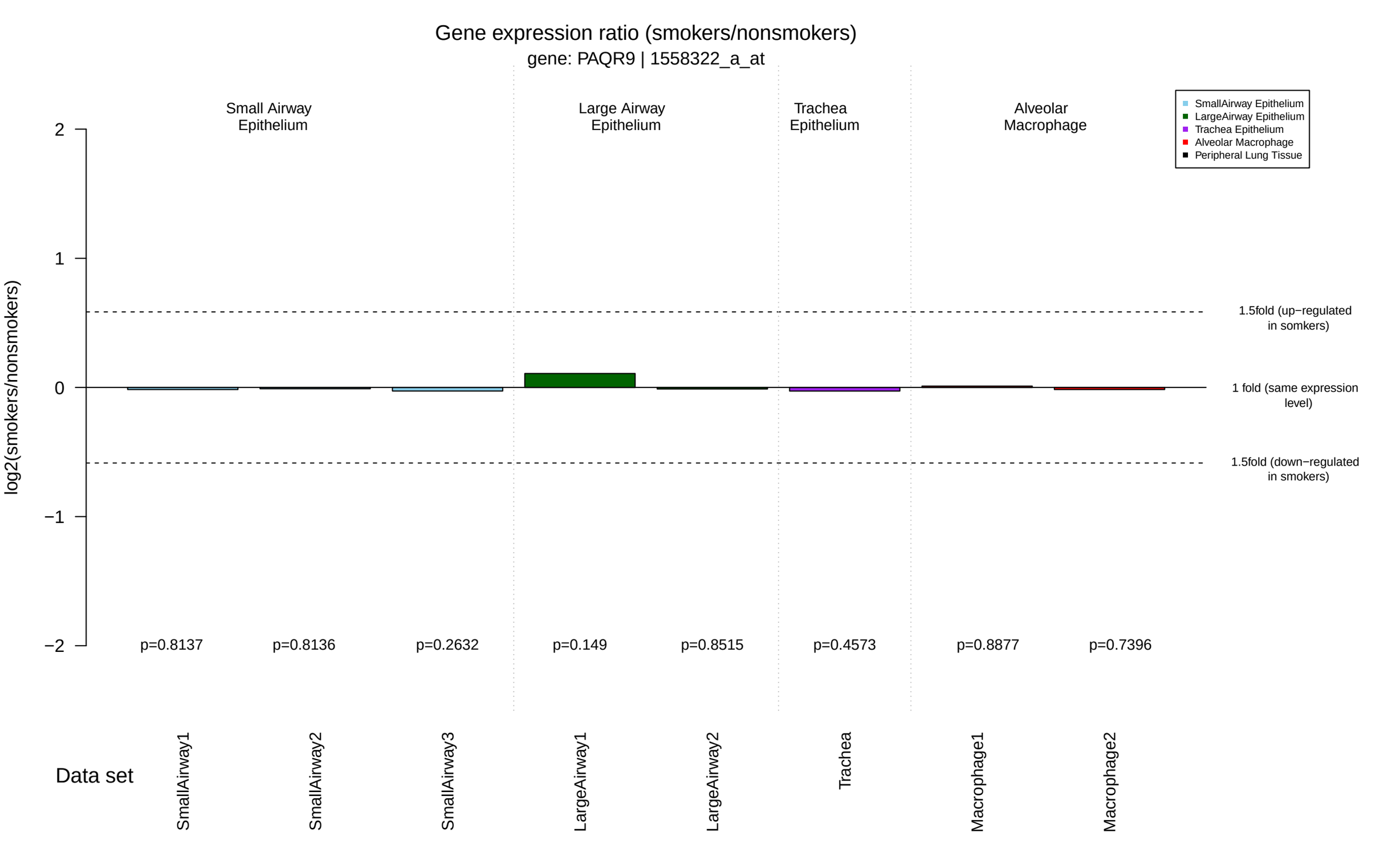
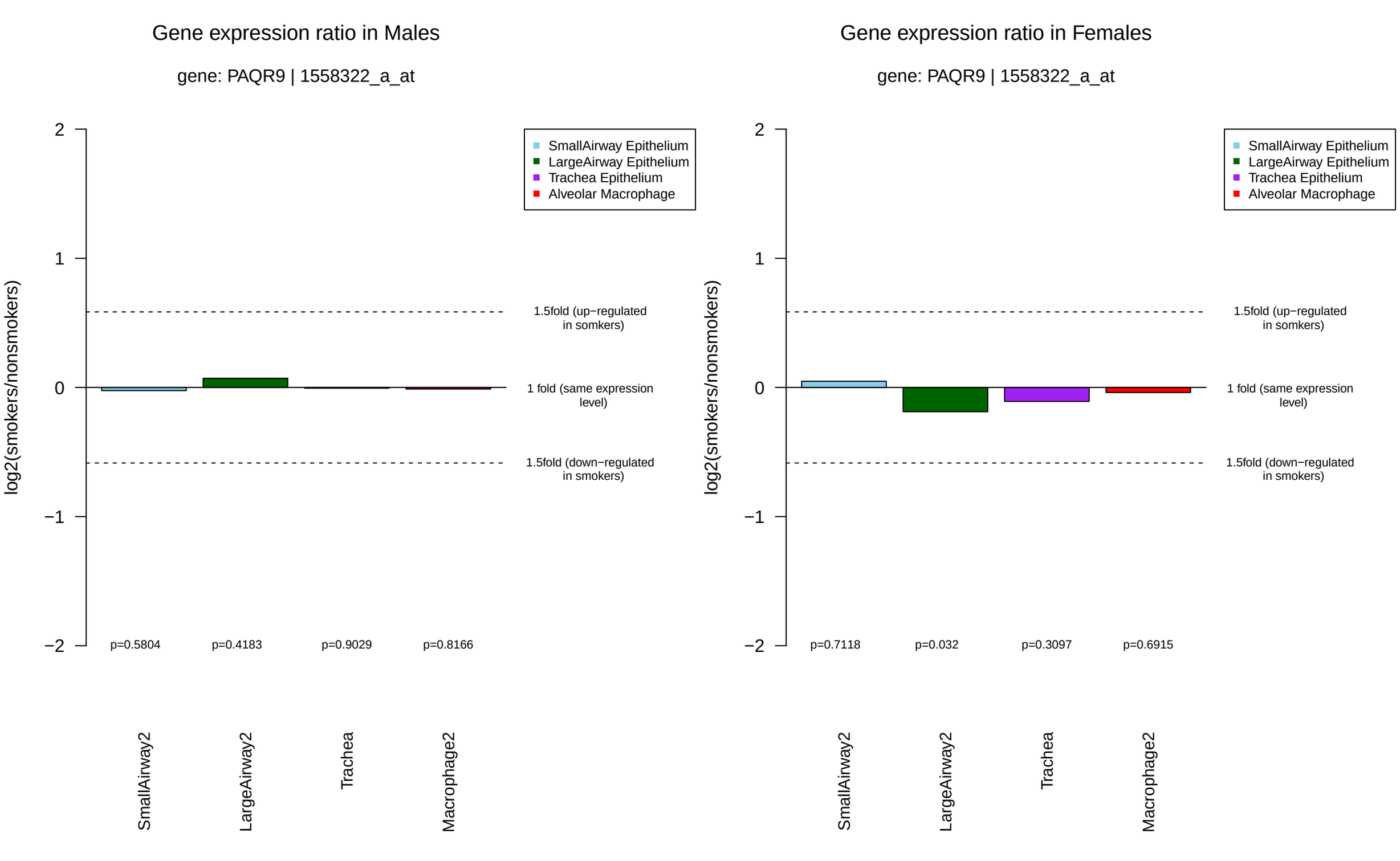
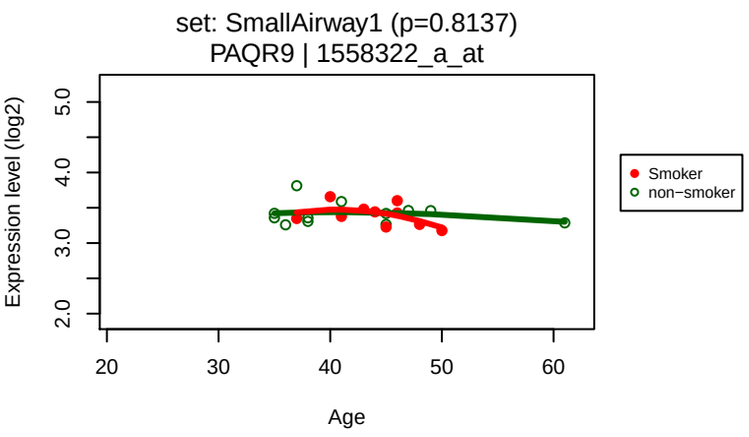
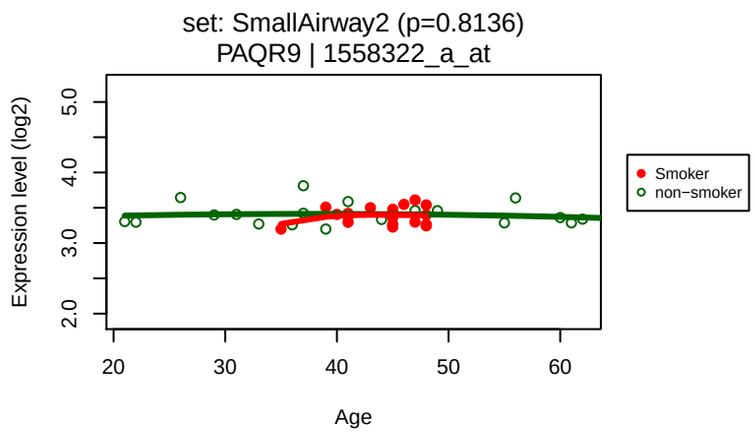
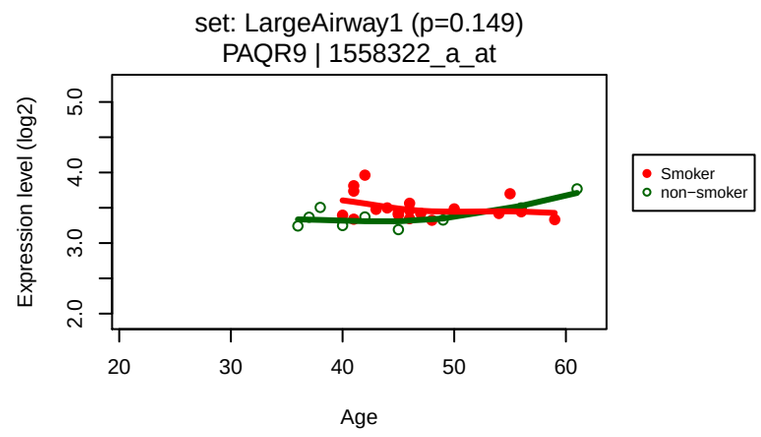

Gene expression of smokers and nonsmokers
gene: PAQR9 | 1558322_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8137
p=0.8136
p=0.2632
p=0.149
p=0.8515
p=0.4573
p=0.8877
p=0.7396
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

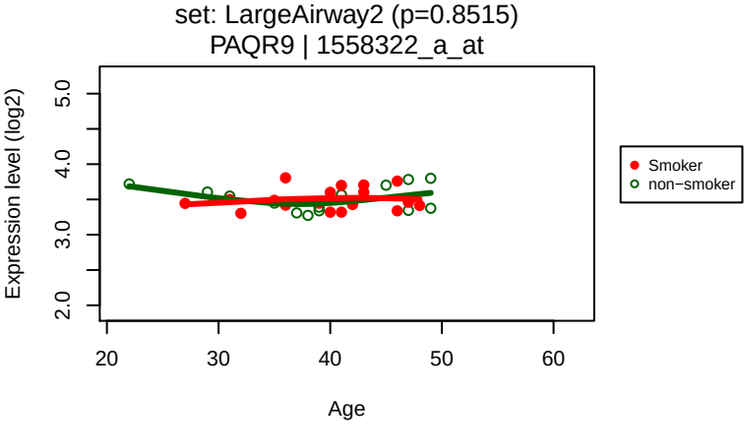

Gene expression ratio (smokers/nonsmokers)
gene: PAQR9 | 1558322_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8137
p=0.8136
p=0.2632
p=0.149
p=0.8515
p=0.4573
p=0.8877
p=0.7396
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

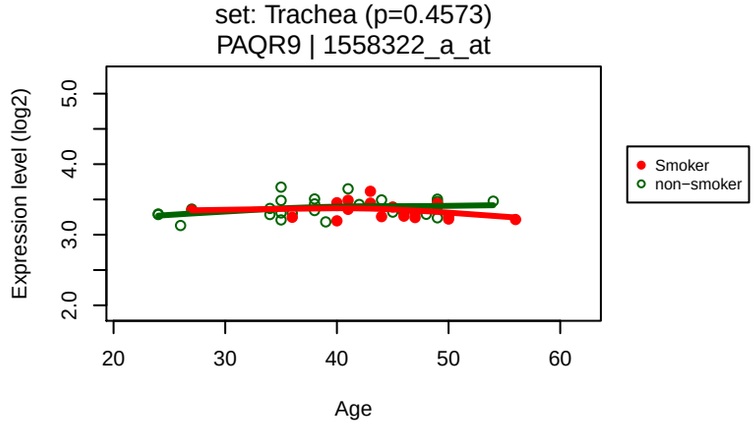

Gene expression ratio in Males
gene: PAQR9 | 1558322_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5804
p=0.4183
p=0.9029
p=0.8166
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAQR9 | 1558322_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7118
p=0.032
p=0.3097
p=0.6915
SmallAirway2
LargeAirway2
Trachea
Macrophage2

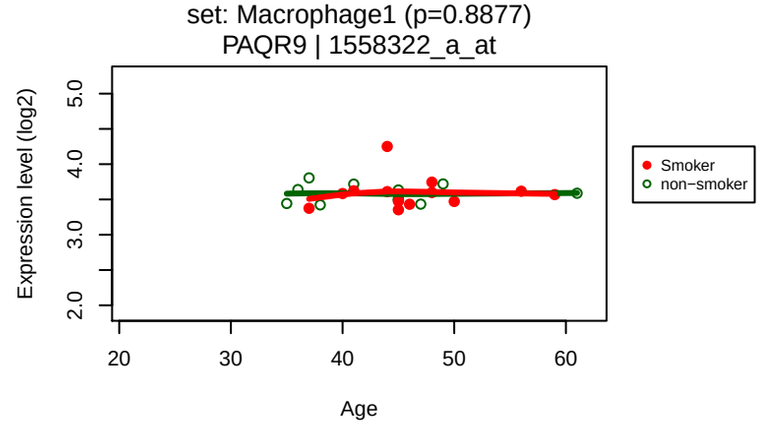

set: SmallAirway1 (p=0.8137)
PAQR9 | 1558322_a_at
Expression level (log2)
Age
Smoker
non−smoker

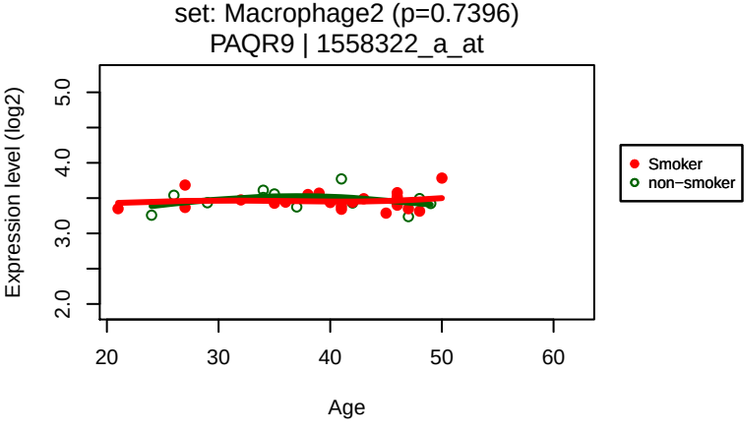

set: SmallAirway2 (p=0.8136)
PAQR9 | 1558322_a_at
Expression level (log2)
Age
Smoker
non−smoker

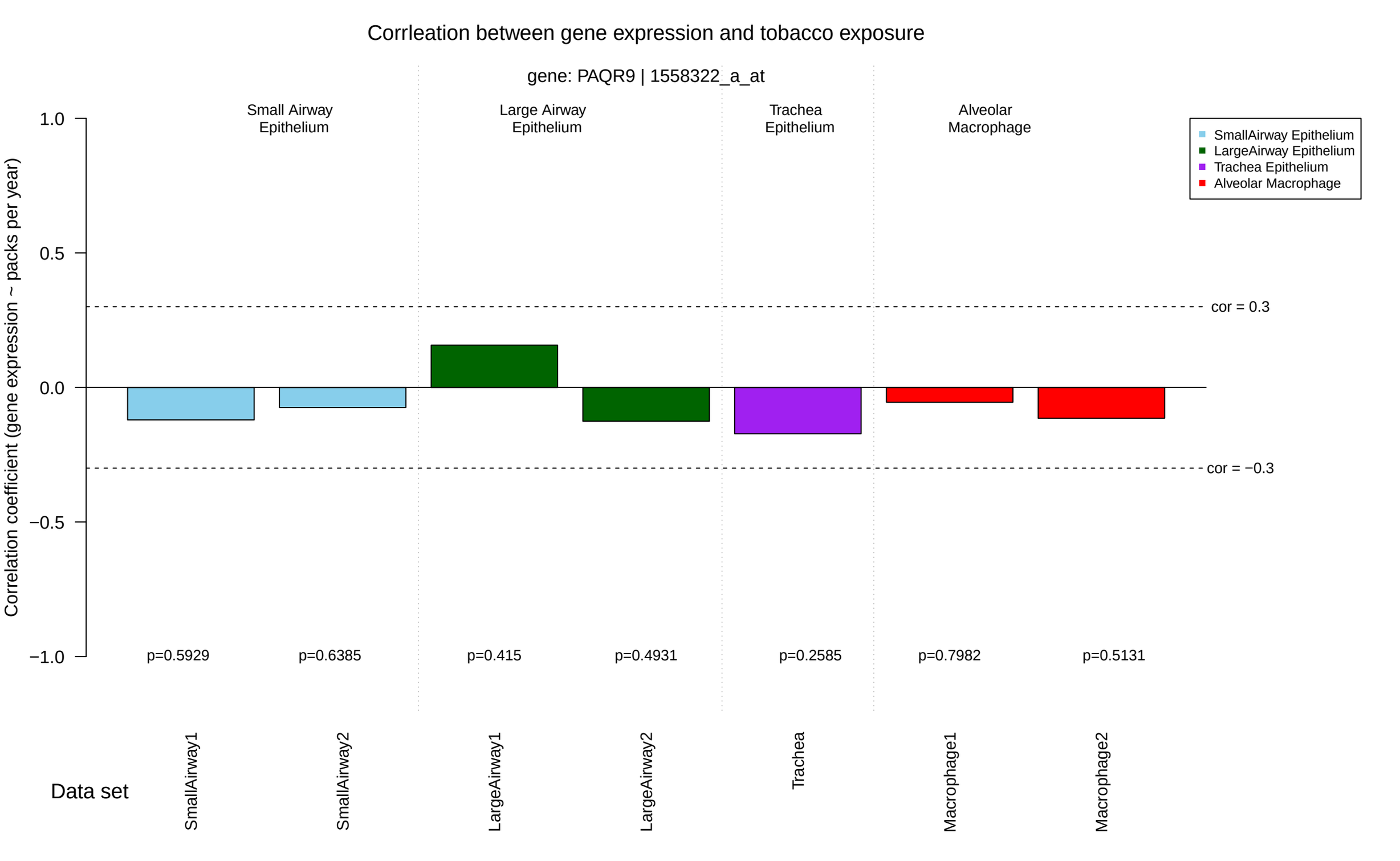

set: LargeAirway1 (p=0.149)
PAQR9 | 1558322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8515)
PAQR9 | 1558322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4573)
PAQR9 | 1558322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8877)
PAQR9 | 1558322_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7396)
PAQR9 | 1558322_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAQR9 | 1558322_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5929
p=0.6385
p=0.415
p=0.4931
p=0.2585
p=0.7982
p=0.5131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2